Introduction to Mobile Computing

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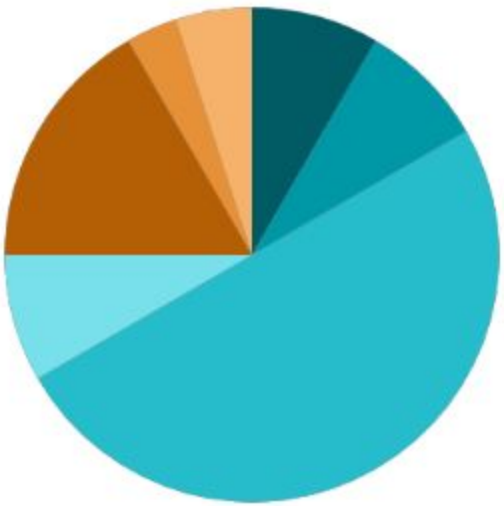
Lecture Structure



05 Minutes	Outline & Objectives
05 Minutes	Recap of Previous Lecture
30 Minutes	Delivery of Actual Content as per Lesson Plan
05 Minutes	Industry Relevance
10 Minutes	Interactive Activities
02 Minutes	Importance of Discipline
03 Minutes	Recording of Attendance

Outline

1. Fundamentals of Mobile Communication
2. Wireless Technology Evolution
3. Mobile Computing Architecture
4. Network Models in Mobile Computing
5. Technologies Enabling Mobility

Objectives

1. Understand the core concepts and importance of mobile communication.

2. Explore the evolution of wireless technologies from 1G to 5G.

3. Learn the architecture and functional components of GSM and mobile computing systems.

4. Familiarize with different network types like MANETs and satellite systems.

5. Introduce key technologies and software models like mobile agents that support mobile computing.

Concepts of Mobile Communication

1. Definition : Mobile communication refers to the ability to send and receive data or voice wirelessly while moving.

2. Key Characteristics : Wireless connectivity, Mobility support, Network availability, Device portability

3. Components : Mobile Devices (smartphones, tablets), Wireless Networks (Wi-Fi, 4G/5G), Base Stations and Access Points, Network Infrastructure (routers, switches, servers)

4. Communication Types :Voice communication Text messaging (SMS), Data communication (Internet access, app usage)Multimedia sharing

Examples : Making a phone call while traveling, Accessing the internet on a mobile device Using WhatsApp, Telegram, or email apps on the go.

Characteristics and Applications of Mobile Communication

- Characteristics: Mobility, Ubiquity, Portability, Wireless, Real-time Access.

- Applications:
- - Personal: Calls, social media.
- - Business: Remote work, meetings.
- - Healthcare: Remote monitoring.
- - Education: E-learning apps.
- - Finance: Mobile banking.

What is Mobile Computing?

- It is not stationary.
- It is using Computing Technology.
- It is a technology that allows any time, anywhere and everywhere computing.
- Allows Transmission of data, voice and video via a computer or any other wireless device.

Uses

- Allows users to perform any task from anywhere using a computing device. [Mobile, tablet, PDA]
- Communication is spread over both wired and wireless media.
- Set of distributed computing systems participate, connect and synchronize through mobile communication protocol.

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2 Concepts

- **Computing:** Processes-related services on remote computer.
- **Communication:** capability to change the location while communicating.
  - Invoke services to remote computer.

Mobile Computing Types/Major Components

- Mobile communication/Infrastructure
- Mobile Hardware
- Mobile Software

Mobile communication/Infrastructure

- Defines a framework responsible for the working of mobile computing technology.
  - Framework includes Wireless networks, Services, Data format, Bandwidth and protocols.
- Can be divide into 4 types.
  - Fixed and Wired
  - Fixed and Wireless
  - Mobile and Wired
  - Mobile and Wireless

Types

- **Fixed and Wired:** In this configuration, devices are fixed at a position and they are connected through a physical link.(2 desktop)
- **Fixed and Wireless:** Devices are fixed at a position, and they are connected through a wireless link. (Wi-Fi routers , communication towers)
- **Mobile and wired:** Some devices are wired and some are mobile.(laptops)
- **Mobile and Wireless:** devices can connect to any network without the use of any Wired device. (wifi dongle)

Mobile Hardware

- Consists of mobile devices or device components that are used to send and receive the service of mobility. (Any computers)

- It is actual program that runs on mobile hardware.It deals with characteristics and requirements of mobile hardware.
- It is the operating system of any device.
- Very crucial component

Mobile computing Characteristics

1. **Ubiquity:** anywhere at any time
   - The ability of a user to perform computation from anywhere and at any time. E.g. : Business Notification
2. **Location Awareness:** Many applications require or value additional information from location-based services.
   - E.g: Traffic control application (GPS) simply tracking location
3. **Adaption:** Ability to adjust to bandwidth fluctuation without inconveniencing the user.
4. **Broadcast:** Efficient delivery of data can be made simultaneously to hundreds of mobile devices.
5. **Personalization:** Services in a mobile environment can be easily personalized according to a user profile.

Drawback/Challenges in Mobile Computing

- Disconnection
- Bandwidth validation
- Heterogeneous Network
- Security Risk

Different generations of Wireless technology,

- **1G (First Generation):**
  - Analog voice communication
  - Introduced in the 1980s
  - Example: AMPS (Advanced Mobile Phone System)
- **2G (Second Generation):**
  - Digital voice communication
  - Introduced in the 1990s
  - Examples: GSM (Global System for Mobile Communications), CDMA (Code Division Multiple Access)
- **3G (Third Generation):**
  - Mobile data communication
  - Introduced in the 2000s
  - Examples: UMTS (Universal Mobile Telecommunications System), EVDO (Evolution-Data Optimized)
- **4G (Fourth Generation):**
  - High-speed mobile data
  - Introduced in the 2010s
  - Examples: LTE (Long Term Evolution), WiMAX (Worldwide Interoperability for Microwave Access)
- **5G (Fifth Generation):**
  - Enhanced data rates, low latency, and improved connectivity
  - Introduced in the late 2010s
  - Examples: mmWave (Millimeter Wave), Sub-6 GHz

3-Tier Architecture of Mobile Computing

An Overview

Introduction to 3-Tier Architecture

- The 3-tier architecture is a client-server architecture that includes three layers: Presentation Tier, Application Tier, and Data Tier.
- Each layer is independent and can be developed and maintained separately.
- Widely used in mobile computing to improve scalability, manageability, and ease of development.

Presentation Tier

- Also known as the User Interface (UI) layer.
- Responsible for displaying information to the user and collecting user input.
- Examples: Mobile apps' graphical user interfaces (GUIs), web browsers.

Application Tier

- Also known as the Logic layer or Business Logic layer.
- Responsible for processing user input, making logical decisions, and performing calculations.
- Acts as an intermediary between the Presentation and Data tiers.
- Examples: Web servers, application servers, business logic components.

Data Tier

- Also known as the Database layer.
- Responsible for storing, retrieving, and managing data.
- Examples: Databases, file systems, cloud storage services.

Benefits of 3-Tier Architecture

- Scalability: Each tier can be scaled independently to accommodate more users or increased load.
- Maintainability: Easier to update and maintain due to the separation of concerns.
- Reusability: Components in each tier can be reused in other applications or projects.
- Flexibility: Allows for the use of different technologies and platforms in each tier.

Example of 3-Tier Architecture in Mobile Computing

- Presentation Tier: Mobile app with a user interface for interacting with users.
- Application Tier: Backend server processing requests and applying business logic.
- Data Tier: Database storing user data, app data, and other relevant information.

Mobile Ad-hoc Networks (MANETs), Mobile Agents, and Global Mobile Satellite Systems

An Overview

Mobile Ad-Hoc Networks (MANETs)

- Definition: Self-configuring, infrastructure-less networks.
- Characteristics: Dynamic, Decentralized, Multi-hop.
- Routing: AODV, DSR, OLSR.
- Applications: Military, Disaster recovery, VANETs.
- Challenges: Security, Routing, Power constraints.

Introduction to MANETs

- Mobile Ad-hoc Networks (MANETs) are self-configuring networks of mobile devices connected by wireless links.
- Each device in a MANET can act as a router, forwarding traffic to other devices.
- MANETs are characterized by their dynamic topologies, decentralized nature, and lack of fixed infrastructure.

Characteristics of MANETs

- Dynamic Topology: Network topology changes frequently due to node mobility.
- Decentralized: No central authority or fixed infrastructure.
- Multi-hop Routing: Data is forwarded through multiple nodes to reach the destination.
- Scalability: Capable of scaling to accommodate a large number of nodes.

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Applications of MANETs

- Military Communication: Secure and reliable communication in battlefield environments.
- Disaster Recovery: Establishing communication networks in areas with damaged infrastructure.
- Vehicular Networks: Enabling communication between vehicles for traffic management and safety.
- Mobile Health: Supporting communication in remote or mobile health monitoring systems.

Introduction to Mobile Agents

- Mobile Agents are software programs that can move autonomously from one node to another in a network.
- They carry code, data, and execution state, allowing them to perform tasks on remote nodes.
- Mobile Agents can reduce network load and improve performance by processing data closer to its source.

Characteristics of Mobile Agents

- Autonomy: Ability to operate without continuous supervision.
- Mobility: Capability to move between nodes in a network.
- Communication: Ability to interact with other agents and system components.
- Intelligence: Equipped with algorithms to perform specific tasks and make decisions.

Applications of Mobile Agents

- **Network Management:** Monitoring and managing network resources efficiently.
- **E-commerce:** Facilitating secure and efficient online transactions.
- **Information Retrieval:** Collecting and filtering information from distributed sources.
- **Distributed Computing:** Performing computations across multiple network nodes.

Introduction to Global Mobile Satellite Systems

- Global Mobile Satellite Systems provide communication services via satellites to users across the globe.
- They enable voice, data, and multimedia communication in remote and underserved areas.
- Key players include Iridium, Globalstar, and Inmarsat.

Characteristics of Global Mobile Satellite Systems

- **Wide Coverage:** Provides communication services to remote and underserved regions.
- **Reliability:** Offers reliable communication services, often as a backup to terrestrial networks.
- **Mobility:** Supports communication for users on the move, such as ships, aircraft, and vehicles.
- **Cost:** Typically higher operational costs compared to terrestrial networks.

Different Generations of Wireless Technology

- 1G: Analog, Voice-only, ~2.4 Kbps.
- 2G: Digital, SMS support, ~64 Kbps.
- 3G: Internet access, Video calls, ~2 Mbps.
- 4G: High-speed data, Streaming, ~1 Gbps.
- 5G: IoT support, Ultra-fast, ~10 Gbps+

Basis of GSM Architecture

- **GSM (Global System for Mobile Communications)** is a **second-generation (2G)** digital cellular network standard developed to replace analog systems (1G).
- It supports voice calls, SMS, and basic data services over mobile networks.

Basis of GSM Architecture

- Mobile Station (MS): Handset + SIM.
- Base Station Subsystem (BSS): BTS & BSC.
- Network Switching Subsystem (NSS): MSC, HLR, VLR, AuC, EIR.
- Operation Support System (OSS): Network management.
- Flow: MS ↔ BTS ↔ BSC ↔ MSC ↔ External Networks.

Main Components of GSM Architecture

- **1. Mobile Station (MS)**
- **Mobile Equipment (ME):** User's device (phone, modem).
- **SIM (Subscriber Identity Module):** Stores **IMSI**, **authentication keys**, and user data.

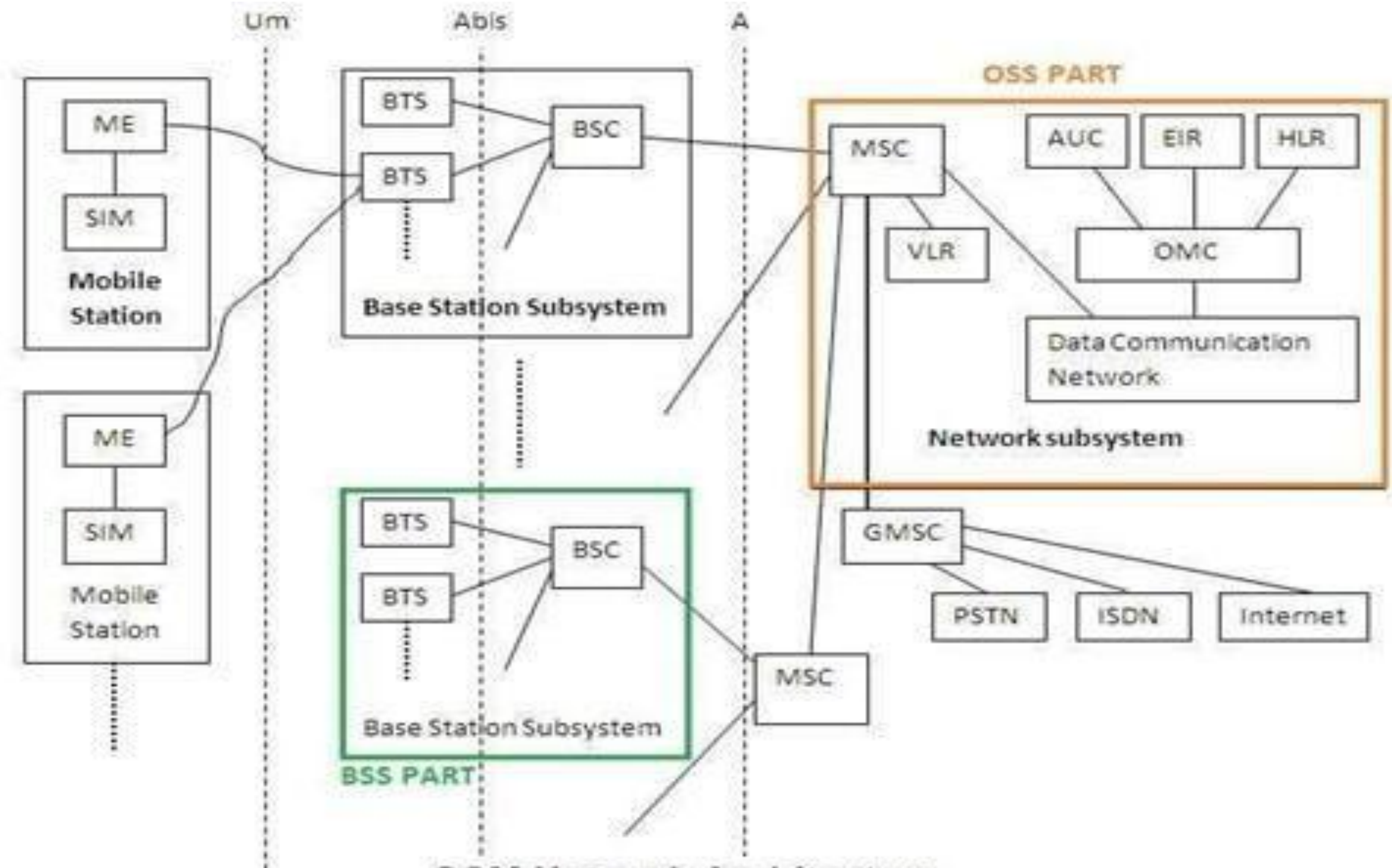
- **2. Base Station Subsystem (BSS)**
- **a. BTS (Base Transceiver Station)**
- Communicates **directly with the mobile device** via radio signals.
- Each BTS covers a **cell**.
- **b. BSC (Base Station Controller)**
- Manages multiple BTSs.
- Handles **handover**, **frequency allocation**, and **call setup**.

Main Components of GSM Architecture

- **3. Network Switching Subsystem (NSS)**
- **a. MSC (Mobile Switching Center)**
- Central node in the NSS.
- Handles **call routing**, **SMS**, and **mobility management**.
- **b. HLR (Home Location Register)**
- Database storing permanent subscriber data like:
  - IMSI, MSISDN (phone number)
  - Services subscribed
  - Last known location (MSC area)
- **c. VLR (Visitor Location Register)**
- Temporary database for users **currently in the area**.
- Helps MSC process calls/SMS by providing local subscriber info.
- **d. AUC (Authentication Center)**
- Stores cryptographic keys used for **authenticating SIM cards**.
- Prevents fraud by verifying user identity.
- **e. EIR (Equipment Identity Register)**
- Keeps a list of device IMEIs:
  - **White List** – allowed
  - **Black List** – blocked/stolen
  - **Gray List** – under observation

Main Components of GSM Architecture

- **4. Operations and Support Subsystem (OSS)**
- Supports **network monitoring**, **fault detection**, and **performance management**.
- Usually integrated with the NSS.



GSM Network Architecture

Summary Table

Component	Full Form	Function
MS	Mobile Station	User's phone & SIM
BTS	Base Transceiver Station	Radio link with phone
BSC	Base Station Controller	Manages multiple BTSs
MSC	Mobile Switching Center	Core switch for calls/SMS
HLR	Home Location Register	Stores permanent user info
VLR	Visitor Location Register	Temporary user info in MSC area
AUC	Authentication Center	Verifies user identity
EIR	Equipment Identity Register	Tracks phone validity

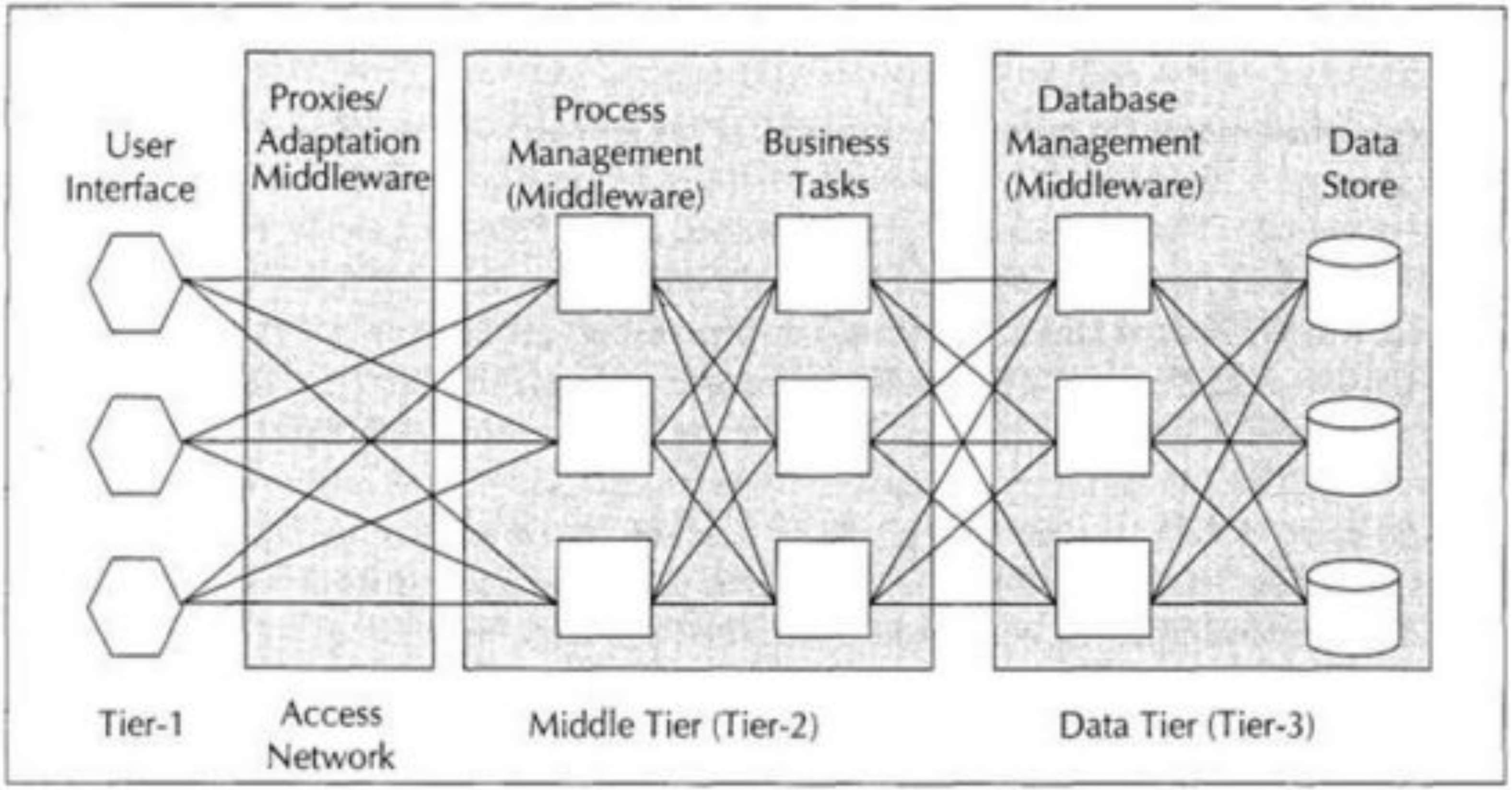
3-Tier Architecture of Mobile Computing

- Presentation Tier: UI on mobile devices.
- Application Tier: Business logic processing.
- Data Tier: Databases and data storage.
- Advantages: Scalable, Secure, Maintainable.

- **1. Presentation Tier (User Interface Layer)**
- **Also called:** Client Tier / Frontend
- **Role:** Interface between the user and the system.
- **Runs on:** Mobile device (smartphone, tablet, PDA, etc.)
- **Components:**
  - GUI (Graphical User Interface)
  - App display and input screens
- **Functions:**
  - Captures user input
  - Displays output
  - Sends/receives data from the application tier
- **Example:** The WhatsApp app screen where users type messages and see chats.

- **2. Application Tier (Business Logic Layer)**
- **Also called:** Middle Tier / Logic Tier
- **Role:** Processes data, applies business logic, and handles communication between the UI and database.
- **Runs on:** Web servers or application servers.
- **Components:**
  - Application programs
  - API handlers
- **Functions:**
  - Validates and processes input
  - Performs core logic
  - Coordinates between client and database
- **Example:** Processing a login request or checking if a user's message should be forwarded to another server.

- **3. Data Tier (Database Layer)**
- **Also called:** Backend / Storage Tier
- **Role:** Stores and retrieves data.
- **Runs on:** Dedicated database servers.
- **Components:**
  - Databases (MySQL, Oracle, MongoDB, etc.)
- **Functions:**
  - Stores user profiles, chat history, app data
  - Executes queries and updates
  - Manages data consistency and integrity
- **Example:** When WhatsApp retrieves old chat history or stores a new image.



User Interface
Proxies/ Adaptation Middleware
Process Management (Middleware)
Business Tasks
Database Management (Middleware)
Data Store
Tier-1
Access Network
Middle Tier (Tier-2)
Data Tier (Tier-3)

Mobile Agents

- Definition: Software that moves across network nodes.
- Characteristics: Autonomous, Mobile, Adaptive.
- Life Cycle: Creation → Migration → Execution → Termination.
- Applications: E-commerce, Info retrieval, System monitoring.
- Advantages: Reduces network load, Works independently.

Global Mobile Satellite Systems (GMSS)

- Definition: Satellite-based communication globally.
- Features: Global coverage, Reliable, Mobile support.
- Providers: Iridium, Inmarsat, Globalstar, Thuraya.
- Applications: Aviation, Maritime, Disaster recovery.
- Advantages: Seamless, Remote access, Backup communication.

Conclusion

- MANETs, Mobile Agents, and Global Mobile Satellite Systems are crucial components of modern mobile communication.
- Each technology offers unique characteristics and applications that enhance communication capabilities.
- Understanding these technologies is essential for leveraging their benefits in various fields.

Industry Relevance Continued....

How Mobile Computing is useful for the industry:

2. Improves Communication and Collaboration

- Enables instant messaging, video calls, and team collaboration platforms.

- Cloud-based document sharing through mobile apps like Microsoft Teams or Google Workspace.

Example: Sales teams can collaborate with product and logistics teams from the field.

Interactive Activity-Mobile Computing :

Live App-Based Demonstration

Objective: Show real-world use of mobile computing

How:

- Students will be asked to use a cloud-based app like **Google Docs**, **Trello**, or **Slack** on their phones.
- They will simulate a situation (e.g., team planning a delivery route or editing a document together remotely).
- They will collaborate live and see real-time changes.

Why it works: Demonstrates mobile collaboration, cloud sync, and real-time computing.

GSFCU & Values : Cultivating a Positive Mindset

Our attitude, not your aptitude, will determine your altitude."
— *Zig Ziglar*

Discipline is more reliable than motivation.
— *Jocko Willink*

Recommended Books